

Verification 12_Loading_Rate

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1 Objective

The purpose of this verification exercise is to demonstrate that the numerical simulator Verification 12_Loading_Rate correctly reproduces the results of Garagash (2021) in the case of no impulse. This is done by digitizing the data of Figures 5 and S5 in Garagash (2021) and comparing them to the simulation results produced by the MATLAB and Python simulators.

2 Results

The results for the aging and slip law simulations can be found in Figures 1 and 2, respectively. In each figure, the no-impulse case of Garagash (2021) is reproduced for both simulators. In addition, a case with a loading rate of $\frac{\partial}{\partial t_s} \left(\frac{\Delta f_0}{b} \right) = 1$ is performed. While these cases cannot be verified against any known solutions, both simulators agree on the result.

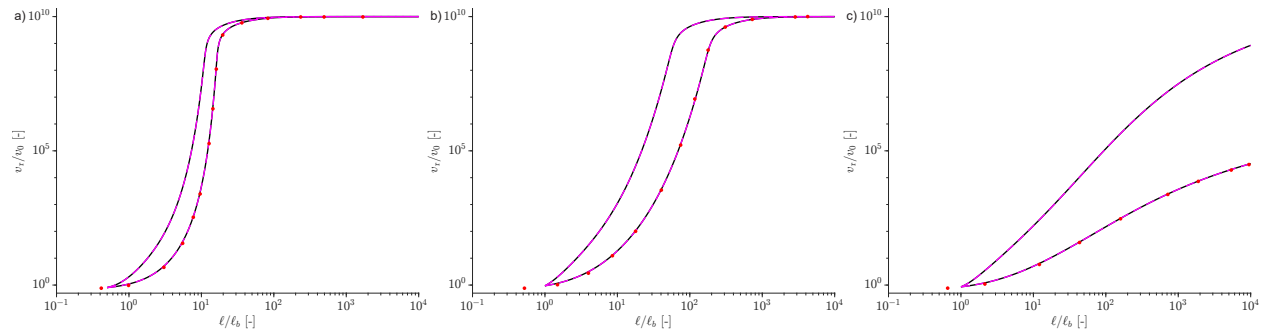


Figure 1: Rupture velocity as function of crack length for MATLAB (black) and Python (dashed pink). Each line corresponds to a different background stress. The digitized results of Garagash (2021) are shown with red dots. This figure corresponds to the aging-law simulations of Figure S5 of Garagash (2021) for $\Delta T=0$, with a/b in the same order as Figure S5 of Garagash (2021). Additionally, one simulation is performed with $\partial_{t_s} (\Delta f_0/b) = 1$. While no third-party solution exists for comparison, both the Python and MATLAB solutions at least agree.

3 Conclusion

The simulators have both been verified to the extent possible.

References

- D. Garagash. Fracture mechanics of rate-and-state faults and fluid injection induced slip. *Philosophical Transactions of the Royal Society A*, 379, 2021. doi: <https://doi.org/10.1098/rsta.2020.0129>.

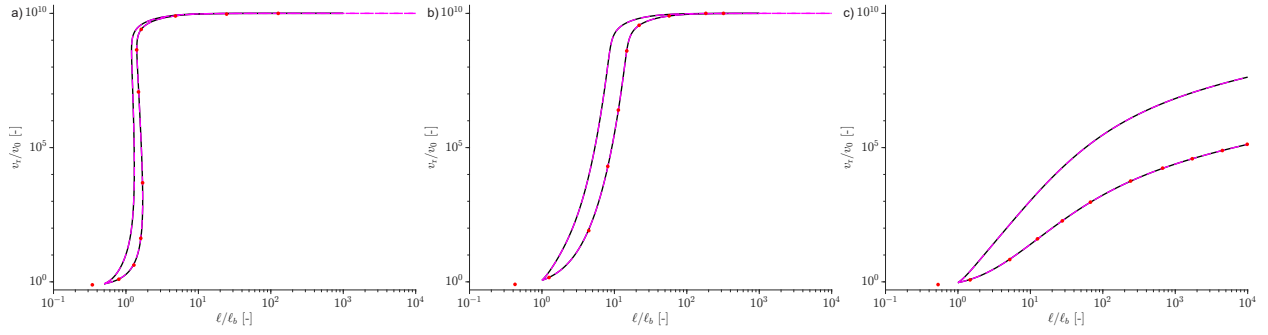


Figure 2: Rupture velocity as function of crack length for MATLAB (black) and Python (dashed pink). Each line corresponds to a different background stress. The digitized results of Garagash (2021) are shown with red dots. This figure corresponds to the slip-law simulations of Figure 5 of Garagash (2021) for $\Delta T=0$, with a/b in the same order as Figure 5 of Garagash (2021). Additionally, one simulation is performed with $\partial_{t_s} (\Delta f_0/b) = 1$. While no third-party solution exists for comparison, both the Python and MATLAB solutions at least agree.